# Solids, liquids and gases

30 questions: 10 multiple choice, 20 structured.

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## 1 · Part 1

Paper 1 & 2 Units [1]

Which row gives the correct units for density and for pressure?

| | density | pressure |
|---|---|---|
| A | $kg/m^3$ | N |
| B | $kg/m^3$ | Pa |
| C | $kg/m^2$ | Pa |
| D | $kg\ m^3$ | J |

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## 2 · Part 1

Paper 1 & 2 Units [1]

One sample of gas has a volume of 450 $cm^3$.

What is this volume in $m^3$?

A $4.5 \times 10^{-2}\ m^3$

B $4.5 \times 10^{-1}\ m^3$

C $4.5\ m^3$

D $4.5 \times 10^{-4}\ m^3$

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## 3 · Part 1

Paper 1 & 2 | Density and pressure | [1]

One metal block has a mass of 540 g and a volume of 200 $cm^3$.

What is the density of the metal?

A. 2700 $kg/m^3$

B. 2.7 $kg/m^3$

C. 27 000 $kg/m^3$

D. 370 $kg/m^3$

## 4 · Part 1

Paper 1 & 2 | Density and pressure | [1]

Which statement about the pressure at a point in a liquid at rest is correct?

A. It acts only downwards.

B. It acts only downwards and sideways.

C. It acts equally in all directions.

D. It is greater downwards than upwards.

## 5 · Part 1

Paper 2 only | Change of state | [1]

Which change of state happens only at the surface of a liquid and can happen at temperatures below the boiling point?

A. evaporation

B. boiling

C. condensation

D. melting

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## 6 · Part 1

Paper 2 only | Change of state | [1]

Which statement describes the specific heat capacity of a substance?

A the energy needed to melt 1 kg of the substance without changing its temperature

B the energy needed to raise the temperature of the substance by 1 °C

C the energy needed to heat 1 kg of the substance from 0 °C to its boiling point

D the energy needed to change the temperature of 1 kg of the substance by 1 °C

## 7 · Part 1

Paper 2 only | Change of state | [1]

One pan contains 0.50 kg of water. The water is heated from 20 °C to 60 °C.

The specific heat capacity of water is 4200 J/kg °C.

change in thermal energy = mass × specific heat capacity × change in temperature

How much thermal energy is transferred to the water?

A 42 000 J

B 84 000 J

C 126 000 J

D 168 000 J

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## 8 · Part 1

Paper 1 & 2 | Ideal gas molecules | [1]

The air inside a freezer is at a temperature of −23 °C.

What is this temperature on the Kelvin scale?

A −250 K

B 23 K

C 250 K

D 296 K

## 9 · Part 1

Paper 1 & 2 | Ideal gas molecules | [1]

Some gas in a sealed container is heated from 20 °C to 80 °C.

Which statement about the molecules of the gas is correct?

A The average speed of the molecules increases.

B All the molecules now move at exactly the same speed.

C Each molecule becomes larger.

D The average speed of the molecules stays the same because the mass of gas does not change.

## 10 · Part 1

Paper 1 & 2 | Ideal gas molecules | [1]

The air trapped in a sealed syringe is slowly compressed to half its original volume. Its temperature does not change.

What happens to the pressure of the gas?

A It halves.

B It stays the same.

C It doubles.

D It increases to four times its original value.

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## 11 · Part 1

Paper 1 & 2 | Density and pressure | [7]

One student investigates the density of a small, irregularly shaped stone.

She pours water into a measuring cylinder. The reading is 50.0 $cm^3$. She then lowers the stone into the water on a thread until it is completely under the water. The new reading is 68.0 $cm^3$.

The mass of the stone is 57.6 g.

(a) State the name of the instrument the student should use to measure the mass of the stone. [1]

(b) Calculate the volume of the stone in $cm^3$. [1]

(c) Calculate the density of the stone in $kg/m^3$. [3]

(d) Describe two precautions the student should take to obtain an accurate value for the volume of the stone. [2]

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## 12 · Part 1

Paper 1 & 2 | Density and pressure | [8]

One wooden crate has a weight of 1800 N. It rests on a flat floor. The base of the crate is a rectangle 1.2 m long and 0.50 m wide.

(a) State the equation linking pressure, force and area. [1]

(b) Calculate the pressure that the crate exerts on the floor. [3]

(c) The crate is lifted onto four small feet. Each foot has an area of 0.0050 $m^2$ in contact with the floor. The weight of the feet can be ignored.

Calculate the new pressure on the floor. [2]

(d) Explain why the crate on its feet is more likely to dent a soft wooden floor than the crate resting on its base. [2]

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## 13 · Part 1 Paper 1 & 2 Density and pressure [6]

During a dive, a diver swims down to a depth of 25 m in the sea.

density of sea water = 1030 kg/m$^3$

gravitational field strength = 10 N/kg

(a) State the equation linking pressure difference, height, density and gravitational field strength. [1]

(b) Calculate the pressure difference between the surface of the sea and a depth of 25 m. [2]

(c) The atmospheric pressure at the surface of the sea is $1.0 \times 10^5$ Pa.

Calculate the total pressure on the diver at a depth of 25 m. [1]

(d) The diver then swims horizontally, staying at a depth of 25 m.

State and explain what happens to the total pressure on the diver. [2]

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## 14 · Part 1

Paper 1 & 2 | Density and pressure | [7]

The roof of a house has a water tank on it. A pipe carries water down from the tank to a tap at ground level. The surface of the water in the tank is 8.0 m above the tap. The tap is closed.

density of water = 1000 kg/m$^3$

gravitational field strength = 10 N/kg

(a) Calculate the pressure difference between the surface of the water in the tank and the closed tap. [2]

(b) The opening of the tap has an area of $2.0 \times 10^{-4}$ m$^2$.

Calculate the force on the closed tap caused by this pressure difference. [3]

(c) The owner replaces the tank with a much wider tank. The depth of water above the tap stays at 8.0 m.

Explain why the pressure difference at the tap does not change. [2]

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## 15 · Part 1

Paper 1 & 2 | Density and pressure | [7]

One student measures the density of cooking oil. He places an empty measuring cylinder on a balance. He adds oil to the cylinder 20 $cm^3$ at a time and records the total mass each time.

| volume of oil / $cm^3$ | mass of cylinder and oil / g |
|---|---|
| 0 | 84.0 |
| 20 | 102.4 |
| 40 | 120.8 |
| 60 | 139.2 |
| 80 | 157.6 |

(a) Identify the independent variable and the dependent variable in this investigation. [2]

(b) Determine the density of the oil in $kg/m^3$. [3]

(c) Explain why the student took a series of readings rather than a single reading. [2]

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## 16 · Part 1

Paper 2 only

Change of state

[6]

Substances can exist as solids, liquids or gases.

(a) Describe the arrangement and motion of the particles in a solid. [2]

(b) Describe the arrangement and motion of the particles in a gas. [2]

(c) Explain why a gas can be squashed into a smaller volume easily, but a solid cannot. [2]

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## 17 · Part 1

Paper 2 only

Change of state

[6]

One student melts some stearic acid in a boiling tube. She then removes the tube from the heat and lets it cool in air. She records the temperature of the stearic acid every minute.

| time / min | temperature / °C |
|---|---|
| 0 | 85 |
| 1 | 77 |
| 2 | 70 |
| 3 | 69 |
| 4 | 69 |
| 5 | 69 |
| 6 | 69 |
| 7 | 62 |
| 8 | 56 |

(a) State the melting point of stearic acid. [1]

(b) Identify, from the table, the two times between which the temperature stays constant. [1]

(c) The stearic acid is transferring energy to the surroundings the whole time.

Explain why its temperature stays constant for part of the time. [3]

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(d) Suggest one improvement to the method that would give a more accurate value for the melting point. [1]

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## 18 · Part 1

Paper 2 only

Change of state

[7]

An aluminium block has a mass of 1.5 kg. It is heated from 18 °C to 42 °C.

specific heat capacity of aluminium = 900 J/kg °C

specific heat capacity of water = 4200 J/kg °C

change in thermal energy = mass × specific heat capacity × change in temperature

(a) Calculate the thermal energy transferred to the aluminium block. [3]

(b) The same amount of thermal energy is transferred to 1.5 kg of water.

Calculate the rise in temperature of the water. [3]

(c) Explain why the rise in temperature of the water is smaller than the rise in temperature of the aluminium. [1]

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## 19 · Part 1

Paper 2 only

Change of state

[7]

One student investigates the specific heat capacity of a metal. She uses a 1.00 kg block of the metal that has two holes drilled in it. An electric heater is placed in one hole and a thermometer in the other. A joulemeter measures the energy supplied to the heater.

These are her results.

| quantity | value |
|---|---|
| starting temperature of the block | 19.0 °C |
| highest temperature of the block | 31.5 °C |
| energy supplied to the heater | 6000 J |

change in thermal energy = mass × specific heat capacity × change in temperature

(a) Calculate the specific heat capacity of the metal. Give the unit. [3]

(b) The accepted value for the specific heat capacity of this metal is 450 J/kg °C.

Explain why the student's value is higher than the accepted value. [2]

(c) Describe two changes to the method that would make the student's value closer to the accepted value. [2]

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## 20 · Part 1

Paper 2 only

Change of state

[6]

One pan of water at 20 °C is placed on a cooker. The water is heated until it reaches 100 °C. The heating continues and the water boils at 100 °C.

(a) State the change of state that happens when the water boils. [1]

(b) Explain, in terms of the energy stored in the water and the particles of the water, what happens while the water is heated from 20 °C to 100 °C, and while it continues to boil at 100 °C. [5]

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## 21 · Part 1

Paper 2 only

Change of state

[7]

One storage heater contains concrete blocks with a total mass of 120 kg. The blocks are heated at night. During the day they cool from 65 °C to 25 °C and release energy into a room.

specific heat capacity of concrete = 880 J/kg °C

specific heat capacity of water = 4200 J/kg °C

change in thermal energy = mass × specific heat capacity × change in temperature

(a) Calculate the energy released by the concrete blocks as they cool from 65 °C to 25 °C. [3]

(b) An engineer suggests storing the energy in a tank of water instead. The water would also cool from 65 °C to 25 °C.

Calculate the mass of water that would release the same amount of energy. [2]

(c) Explain why a material with a high specific heat capacity is useful for storing energy in a heater. [2]

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## 22 · Part 1

Paper 2 only

Change of state

[5]

Some ice cubes are taken out of a freezer at −10 °C and left in a warm room.

(a) The ice first warms up from −10 °C to 0 °C.

Explain why transferring energy to the ice makes its temperature rise. [2]

(b) The ice then melts.

Describe the changes in the arrangement and motion of the particles when the ice melts. [3]

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## 23 · Part 1

Paper 1 & 2 Ideal gas molecules [5]

One balloon is filled with air and tied so that no air can escape.

(a) Explain how the molecules of air inside the balloon exert a pressure on the inside surface of the balloon. [3]

(b) More air is then pumped into a second, identical balloon at the same temperature, until it is the same size as the first balloon.

Explain why the pressure inside the second balloon is greater. [2]

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## 24 · Part 1

Paper 1 & 2 Ideal gas molecules [5]

(a) State the value of absolute zero in degrees Celsius. [1]

(b) Explain why there is a lowest possible temperature. [2]

(c) Liquid nitrogen boils at 77 K.

Calculate this temperature in °C. [1]

(d) The temperature of a classroom is 20 °C.

Calculate this temperature in K. [1]

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## 25 · Part 1 Paper 1 & 2 Ideal gas molecules [7]

One sealed metal can contains gas at a pressure of $2.0 \times 10^5$ Pa when its temperature is 27 °C. The can is heated until the gas reaches 177 °C.

For a fixed mass of gas at constant volume:

p1 / T1 = p2 / T2

(a) Convert 27 °C and 177 °C to kelvin. [2]

(b) Calculate the pressure of the gas at 177 °C. [2]

(c) Give a reason why the volume of the gas can be treated as constant. [1]

(d) The can has a label that says "Do not heat above 50 °C".

Explain why. [2]

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## 26 · Part 1

Paper 1 & 2 Ideal gas molecules [6]

One syringe contains 60 $cm^3$ of air at a pressure of $1.0 \times 10^5$ Pa. The end of the syringe is sealed. The plunger is pushed in slowly until the volume of the air is 24 $cm^3$. The temperature of the air stays the same.

For a fixed mass of gas at constant temperature:

p1V1 = p2V2

(a) Explain why the plunger is pushed in slowly. [1]

(b) Calculate the pressure of the air when its volume is 24 $cm^3$. [3]

(c) The plunger is then pulled out slowly until the pressure of the air is $0.80 \times 10^5$ Pa.

Calculate the new volume of the air. [2]

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## 27 · Part 1

Paper 1 & 2 | Ideal gas molecules | [7]

Some gas is trapped in a cylinder by a piston. No gas can enter or leave the cylinder.

(a) The piston is pushed in so that the volume of the gas decreases. The temperature of the gas stays the same.

Explain, in terms of the molecules, why the pressure of the gas increases. [4]

(b) The piston is then locked so that the volume cannot change, and the gas is heated.

Explain, in terms of the molecules, why the pressure of the gas increases. [3]

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## 28 · Part 1

Paper 1 & 2 | Ideal gas molecules | [7]

One student investigates how the pressure of a gas depends on its temperature. A sealed glass flask of air is placed in a water bath and connected to a pressure gauge. The student heats the water and records the pressure at different temperatures.

| temperature / °C | pressure / kPa |
|---|---|
| 0 | 100 |
| 20 | 107 |
| 40 | 115 |
| 60 | 122 |
| 80 | 129 |

For a fixed mass of gas at constant volume:

p1 / T1 = p2 / T2

(a) Identify one variable that the student must keep constant. [1]

(b) Show that the pressure is proportional to the Kelvin temperature. Use the readings at 0 °C and at 80 °C. [3]

(c) Predict the pressure of the air at 100 °C. [2]

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(d) Explain why the student waits a few minutes at each temperature before reading the pressure gauge. [1]

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## 29 · Part 1

Paper 1 & 2 | Ideal gas molecules | [6]

(a) State the relationship between the Kelvin temperature of a gas and the average kinetic energy of its molecules. [1]

(b) A sample of helium is heated from 27 °C to 327 °C.

Determine the factor by which the average kinetic energy of the helium molecules increases. [3]

(c) Another sample of helium is heated from 20 °C to 40 °C. One student says, "The Celsius temperature has doubled, so the average kinetic energy of the molecules has doubled."

Explain why the student is wrong. [2]

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## 30 · Part 1 Paper 1 & 2 Ideal gas molecules [6]

One small bubble of air is released at the bottom of a lake, 15 m below the surface. At the bottom, the volume of the bubble is 2.0 $cm^3$. The temperature of the water is the same at all depths.

density of water = 1000 $kg/m^3$

gravitational field strength = 10 N/kg

atmospheric pressure at the surface = $1.0 \times 10^5$ Pa

For a fixed mass of gas at constant temperature:

p1V1 = p2V2

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(a) Calculate the pressure difference caused by 15 m of water. [2]

(b) Calculate the total pressure on the bubble at the bottom of the lake. [1]

(c) Calculate the volume of the bubble when it reaches the surface. [2]

(d) Explain why the temperature of the water must be the same at all depths for your calculation in (c) to be valid. [1]

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